

DRY-RUN GUIDE | 05_pattern_55555.cpp
Source: cpp\2nd_belt\05_pattern_55555.cpp

CORE BEHAVIOR

For each descending i from N to 1, prints i exactly i times on a line.

REPRESENTATIVE INPUT / CALL

Input: '5'.

STEP TRACE

$i=5$ emits five 5s; then 4 emits four 4s; continue through one 1.

EXPECTED OUTPUT / RETURN

55555 / 4444 / 333 / 22 / 1 (each on its own line).

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program to generate and print the given number pattern.
004 |
005 | For N = 5:
006 | 55555
007 | 4444
008 | 333
009 | 22
010 | 1
011 |
012 | Input Format:
013 | The input consists of a single integer N.
014 |
015 | Output Format:
016 | Print the number pattern as described.
017 | */
018 |
019 | #include <iostream>
020 | using namespace std;
021 |
022 | int main() {
023 |     int n;
024 |     cin >> n;
025 |
026 |     for (int i = n; i >= 1; i--) {
027 |         for (int j = 1; j <= i; j++) {
028 |             cout << i;
029 |         }
030 |         cout << '\n';
031 |     }
032 |
033 |     return 0;
034 | }
035 |
```